

Deploy the RAG Knowledge Base

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Course Content

- Master the local deployment, debugging, and testing processes and methods for RAG knowledge bases
- Master methods for expanding RAG knowledge bases based on your own task scenarios

Tip

- RAG knowledge bases help general-purpose AI large models provide reference knowledge in vertical domains, avoiding AI large model hallucination responses and improving the model's response capability in vertical domain knowledge.
- RAG knowledge bases can help robots quickly expand generalization capabilities across different task scenarios.

Start Dify

Open the terminal. The yellow text shows the current virtual machine's IP address, or enter `ifconfig` in the terminal to check.

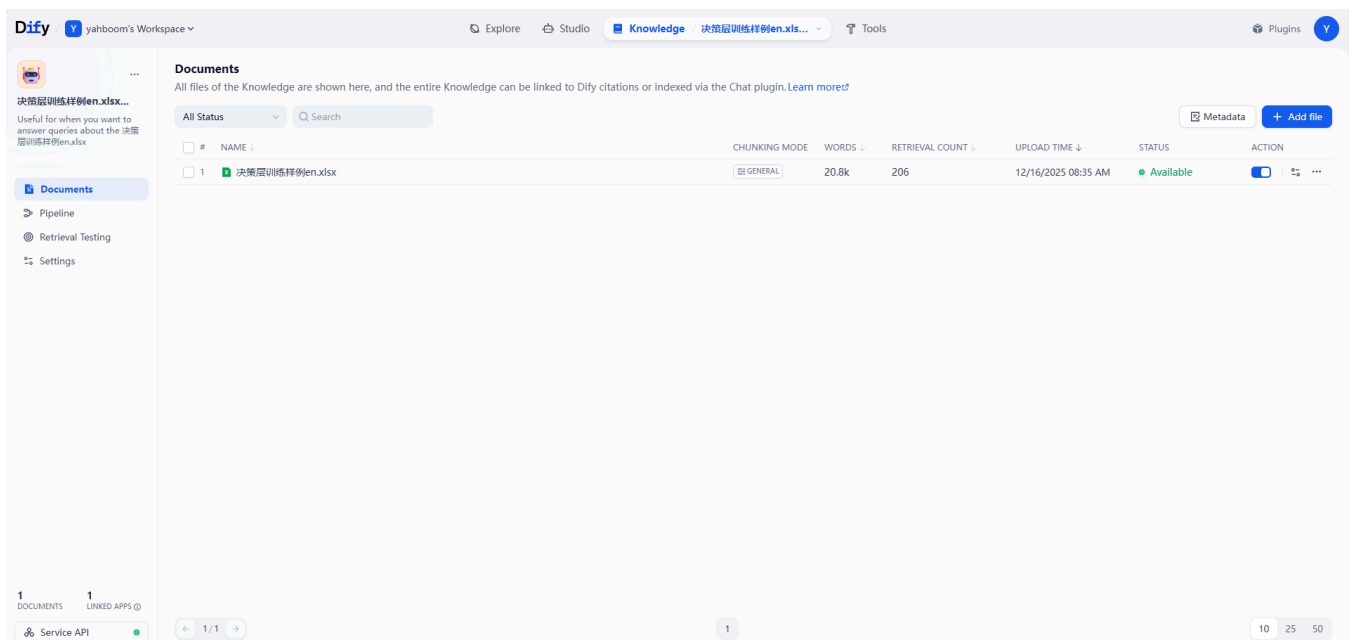
```
yahboom@yahboom-virtual-machine: ~
yahboom@yahboom-virtual-machine: ~ 80x24
=====
IP : 192.168.11.238
DomainID : 30
=====
yahboom@yahboom-virtual-machine:~$
```

View Preset Knowledge Bases

- Click the Knowledge page on the homepage. Dify has 2 preset RAG knowledge bases at the factory. The content is the same, just in different languages.

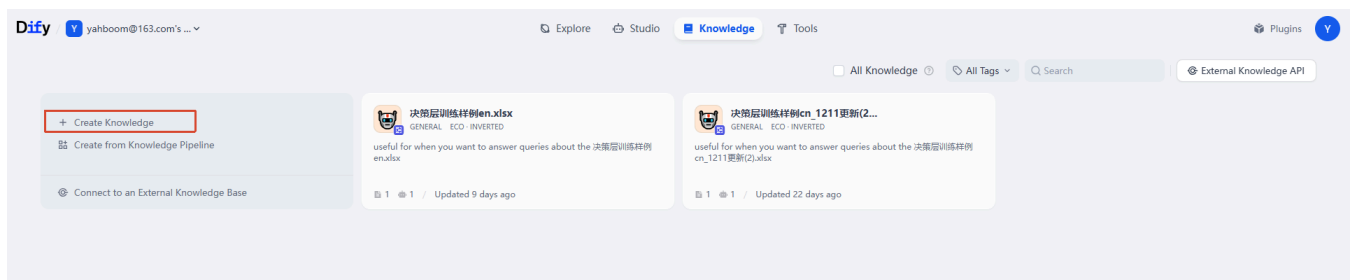
💡 Tip

- The preset knowledge bases here provide some training samples for task scenarios to help AI models quickly master related skills.
- Open a knowledge base. Inside is the `Decision Layer Sample Library` preset file, where:
 - Decision Layer Sample Library: Stores preset reference samples related to specific task scenarios.

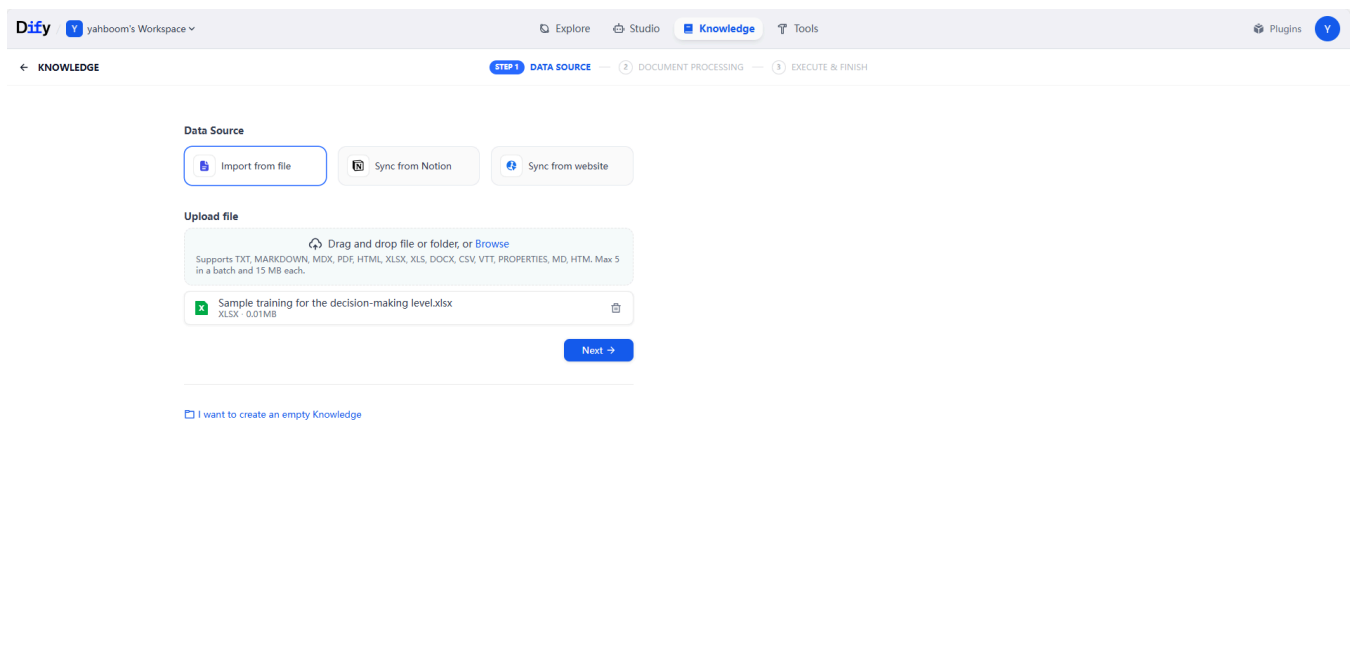


Expand RAG Knowledge Base

- If you need to expand a new knowledge base, click Create Knowledge Base.



- Here, taking importing local data as an example,
- Click "Import from file" -> Browse -> Next



- Then enter the knowledge base configuration page. Click the preview block to check the chunking effect of the file. Here, select "Economical" for the indexing mode.

💡 Tip

- For beginners, it is recommended to first use the economic mode for learning and testing. The differences between the two indexing modes are:
 - Economic: Retrieves content from the knowledge base through **keywords**. Cannot perform extended semantic similarity retrieval. The retrieval method for knowledge fragments is relatively rigid.
 - High-Quality mode: Requires embedding models to consume additional tokens and needs a rerank model. Can achieve more accurate similar semantic fragment retrieval.
- The factory default knowledge base mode is high-quality mode knowledge base.

Economic Mode Knowledge Base

- After selecting the following configuration, click "Save & Process".

The screenshot shows the Dify Knowledge configuration interface. The 'DOCUMENT PROCESSING' step is active. The configuration includes:

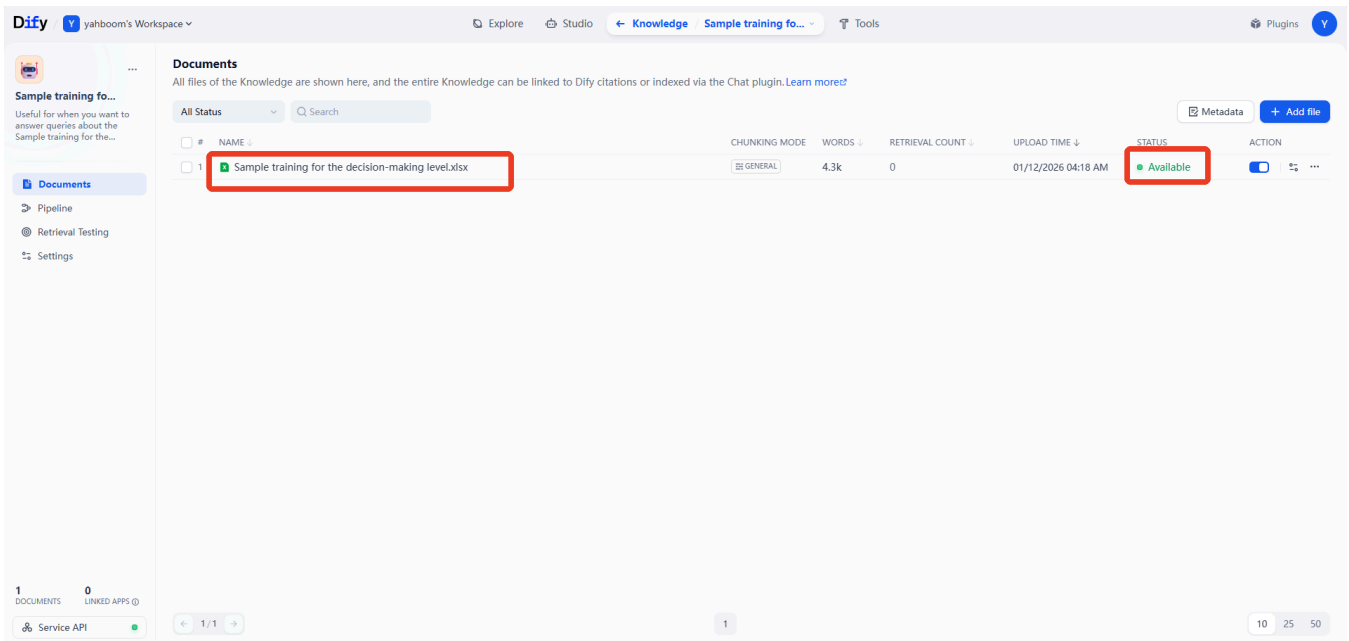
- Delimiter:** \n\n
- Maximum chunk length:** 1024 characters
- Chunk overlap:** 50 characters
- Text Pre-processing Rules:**
 - ☒ Replace consecutive spaces, newlines and tabs
 - ☐ Delete all URLs and email addresses
 - ☐ Chunk using Q&A format in English
- Preview Chunk:** A button to preview the chunk processing results.
- Parent-child:** A section explaining the parent-child mode.
- Index Method:** A section with two options:
 - High Quality:** Recommended, calling the embedding model for more precise retrieval.
 - Economical:** Using 10 keywords per chunk for retrieval, no tokens are consumed at the expense of reduced retrieval accuracy. This option is highlighted with a red box.
- Retrieval Setting:** A section with a slider for 'Top K' set to 3.
- Preview:** A section showing the processed chunks and their corresponding queries. The queries are highlighted with a red box.

- Then wait for the embedding to complete and click "Go to the document".

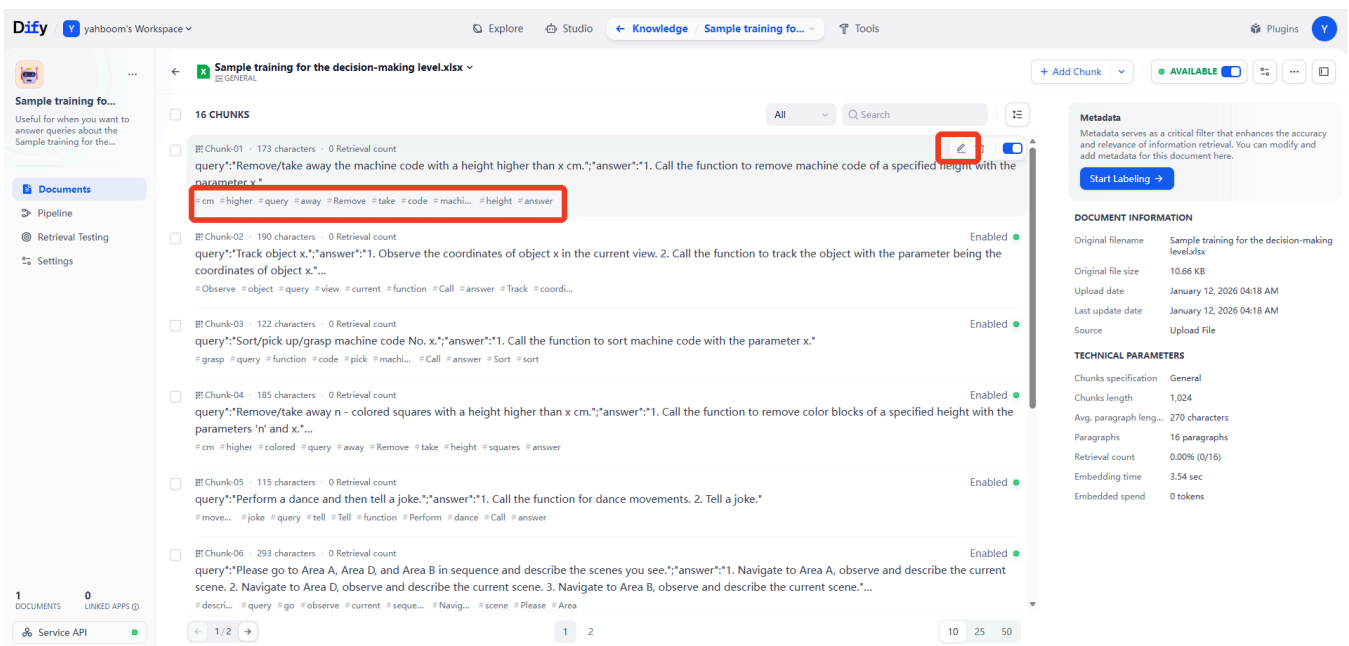
The screenshot shows the Dify Knowledge configuration interface after the 'EXECUTE & FINISH' step. The configuration is complete, and the status is 'EMBEDDING COMPLETED'. The configuration includes:

- Knowledge name:** Sample training fo...
- EMBEDDING COMPLETED:** A section showing the completion status of the embedding process.
- Chunking Setting:** Custom
- Maximum Chunk Length:** 1024
- Text Preprocessing Rules:** Replace consecutive spaces, newlines and tabs
- Index Method:** Economical
- Retrieval Setting:** Inverted Index
- Access the API:** A button to access the API.
- Go to document:** A button to go to the document, highlighted with a red box.
- What's next:** A section providing guidance on how to use the knowledge base.

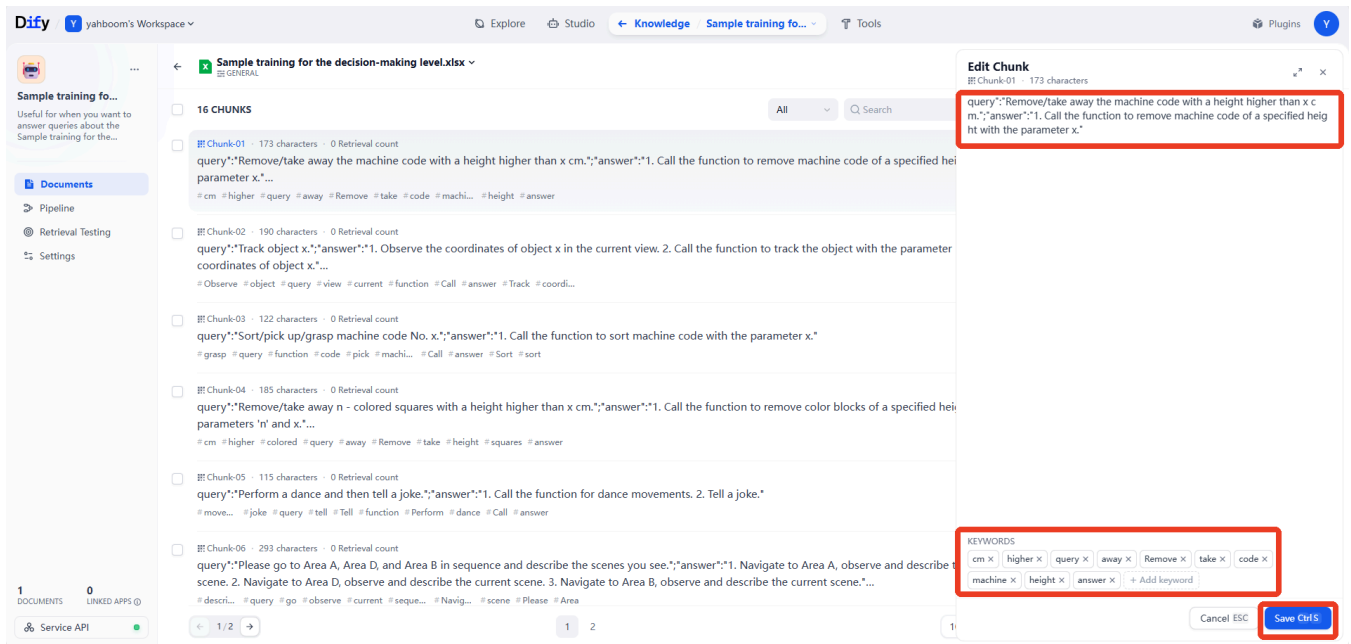
- When the knowledge base is normal, the status will show as available. Then click the knowledge base's file.



- Then we can see the knowledge base fragments that have been segmented. Each fragment has small text below indicating the automatically segmented keywords for that chunk (only the economic mode has this).

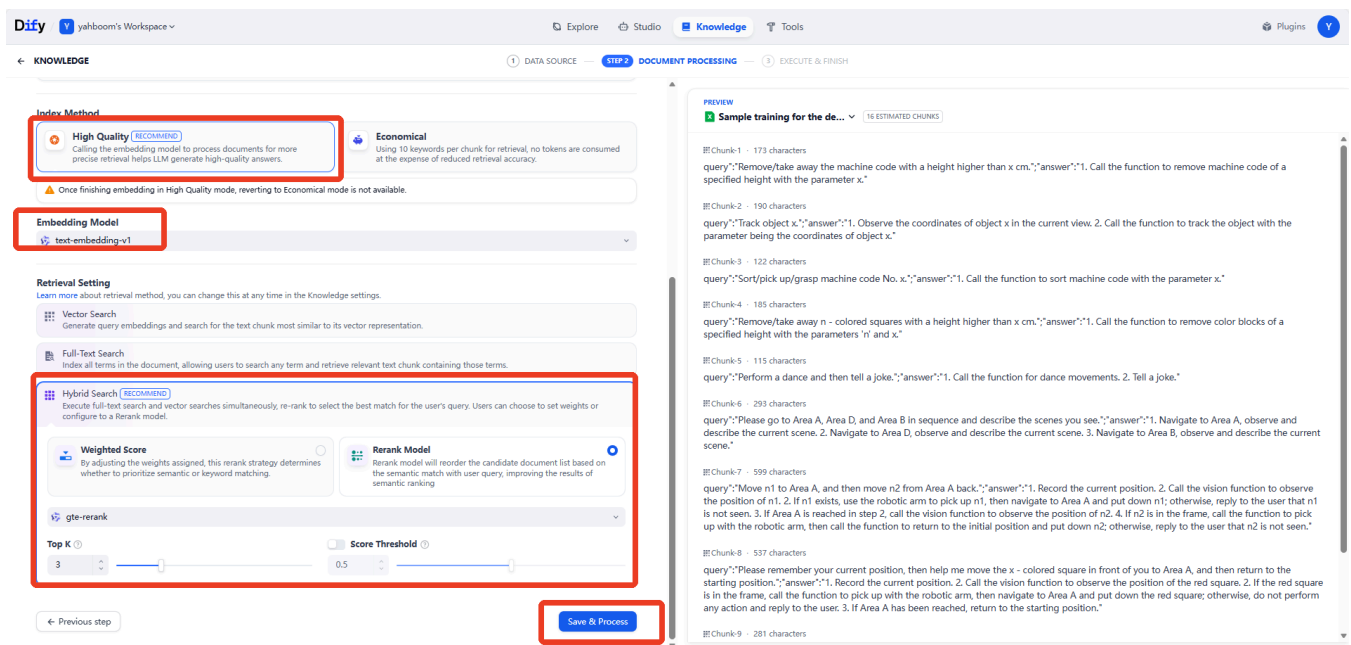


- If the keywords do not accurately describe the knowledge fragment, click "Edit" on the right side of the fragment to edit the content or keywords of that fragment. The image below shows the modified keywords, then click save.



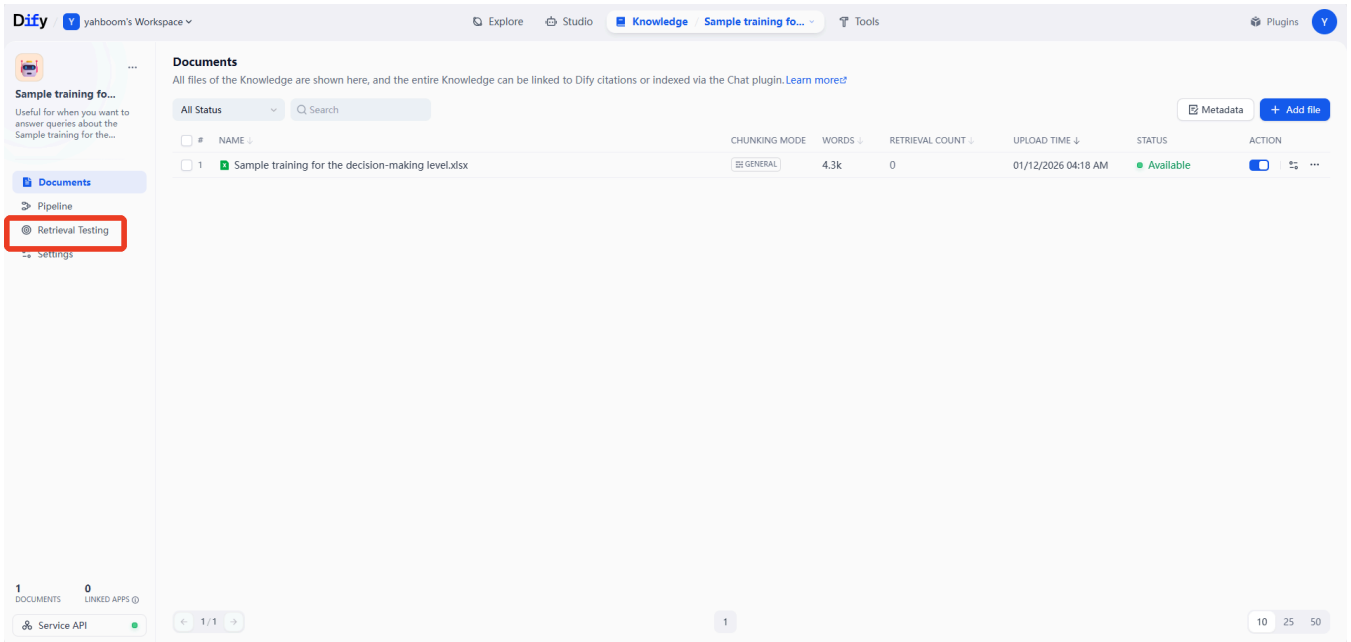
High-Quality Mode Knowledge Base

- If you need to use the high-quality knowledge base later, refer to this section's tutorial.
- The knowledge base creation and file import are the same as above.
- Here, select "High-Quality" for the indexing method. Choose any for the retrieval method. Here we take "Hybrid Retrieval" as an example. Finally, save and process.

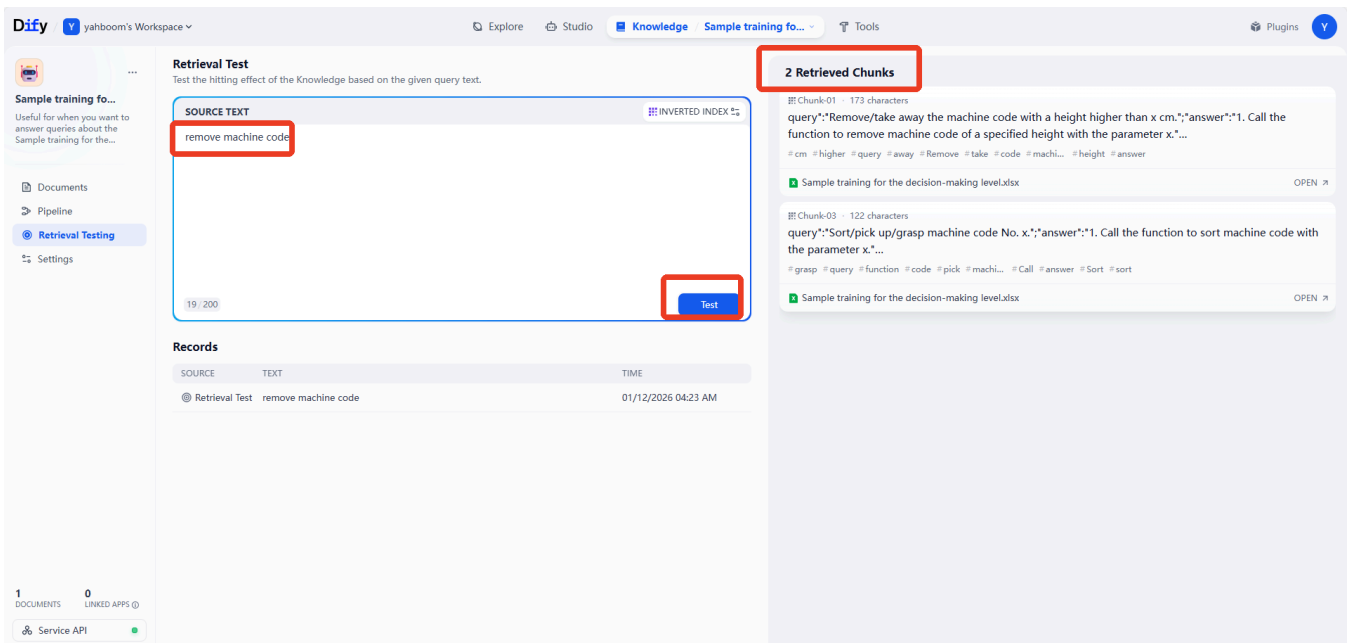


Recall Test

- Recall testing tests the actual effectiveness of retrieving relevant knowledge snippets from the knowledge base based on the input, helping to optimize the AI model's response performance.
- After opening a knowledge base, click on "Recall Test" on the left side.



- Enter the test content in the source text (simulating user input during actual use), and then click "Test".
- The retrieved paragraphs and the knowledge base related to the input content will appear on the right. The knowledge base tested here is the economic model knowledge base, which retrieves information based on keywords.



- If it's a high-quality mode knowledge base, the retrieved snippets will have a `SCORE` rating. A higher score indicates a higher relevance between the snippet and the input content. High-quality mode knowledge bases can perform associative retrieval of similar semantics, but also consume tokens.

Diffy | yahboom's Workspace

Explore Studio Knowledge Sample training fo... Tools

Sample training fo...
Useful for when you want to answer queries about the Sample training for the...

Documents Pipeline Retrieval Testing Settings

1 DOCUMENTS 0 LINKED APPS
Service API

Retrieval Test

Test the hitting effect of the Knowledge based on the given query text.

HYBRID SEARCH

SOURCE TEXT

remove machine code

19 / 200

Test

Records

SOURCE	TEXT	TIME
@ Retrieval Test	remove machine code	01/12/2026 04:25 AM

3 Retrieved Chunks

Chunk-01 173 characters
query: "Remove/take away the machine code with a height higher than x cm."; answer: "1. Call the function to remove machine code of a specified height with the parameter x." ...
SCORE 0.62

Sample training for the decision-making level.xlsx
OPEN

Chunk-03 122 characters
query: "Sort/pick up/grasp machine code No. x."; answer: "1. Call the function to sort machine code with the parameter x." ...
SCORE 0.19

Sample training for the decision-making level.xlsx
OPEN

Chunk-04 185 characters
query: "Remove/take away n - colored squares with a height higher than x cm."; answer: "1. Call the function to remove color blocks of a specified height with the parameters 'n' and 'x'." ...
SCORE 0.05

Sample training for the decision-making level.xlsx
OPEN

This case is for reference only. The USB camera has no depth data. Removing height anomaly codes will cause the machine to be unable to run.